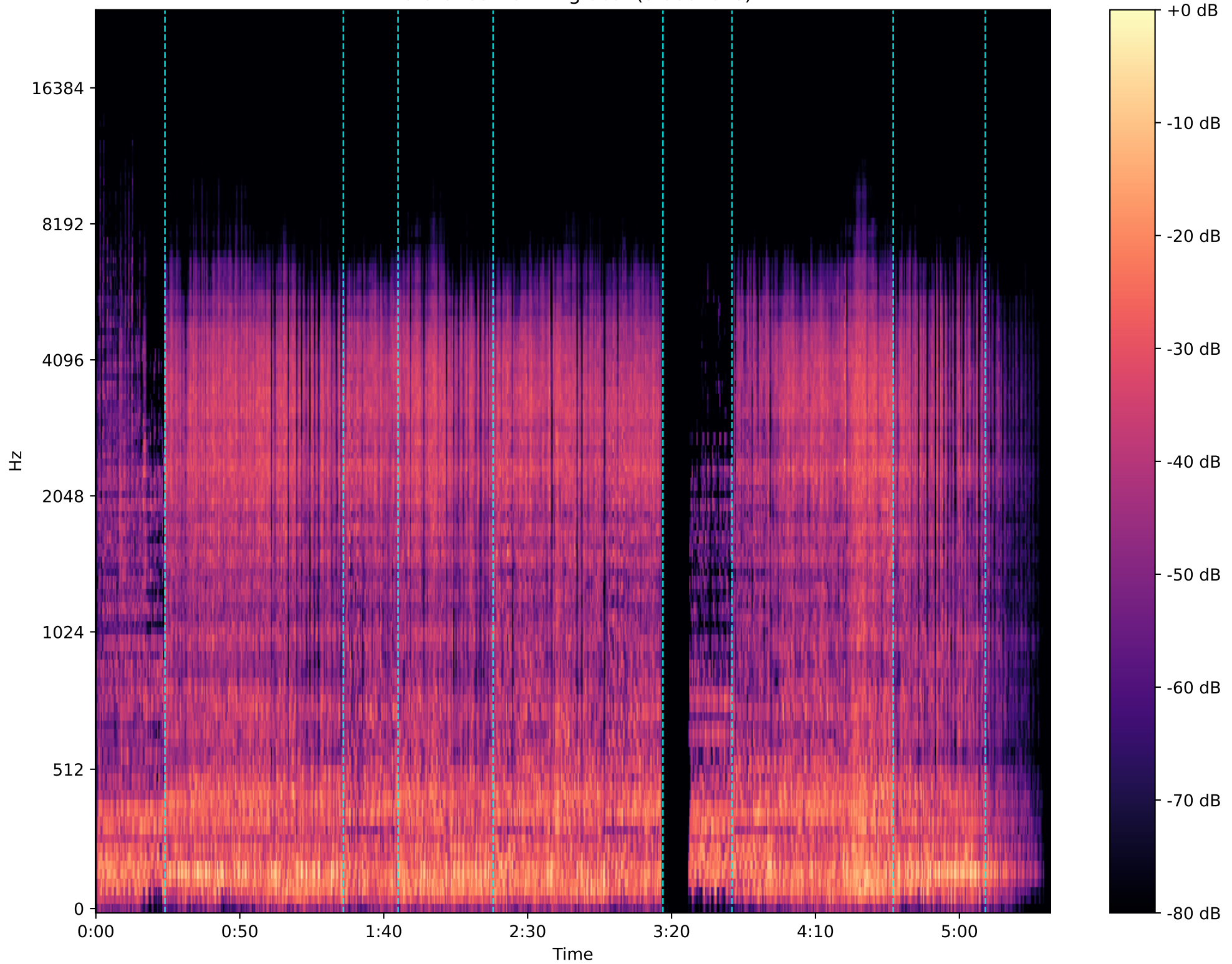


Tone Analysis — reference.wav

Source	/Users/joao.faria/Projetos/github.com/jpfaria/tone-analyzer/tones/metallica-e
SHA-256	afa5dc57e52a45d414d9f3b04334ba439363f7e0cf0518c22d9911b06942c86a
Sample rate	48000 Hz
Channels	2
Duration	331.56 s
Schema	v4
LUFS (int.)	-24.6 LUFS
Peak	-6.6 dB
Stereo width	—
Sections	9 (see following pages)

reference.wav — global (9 sections)



section_0 · 0.00 s - 24.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = rhythm

Loudness RMS = -28.4 dB Peak = -14.8 dB Crest = 13.6 dB

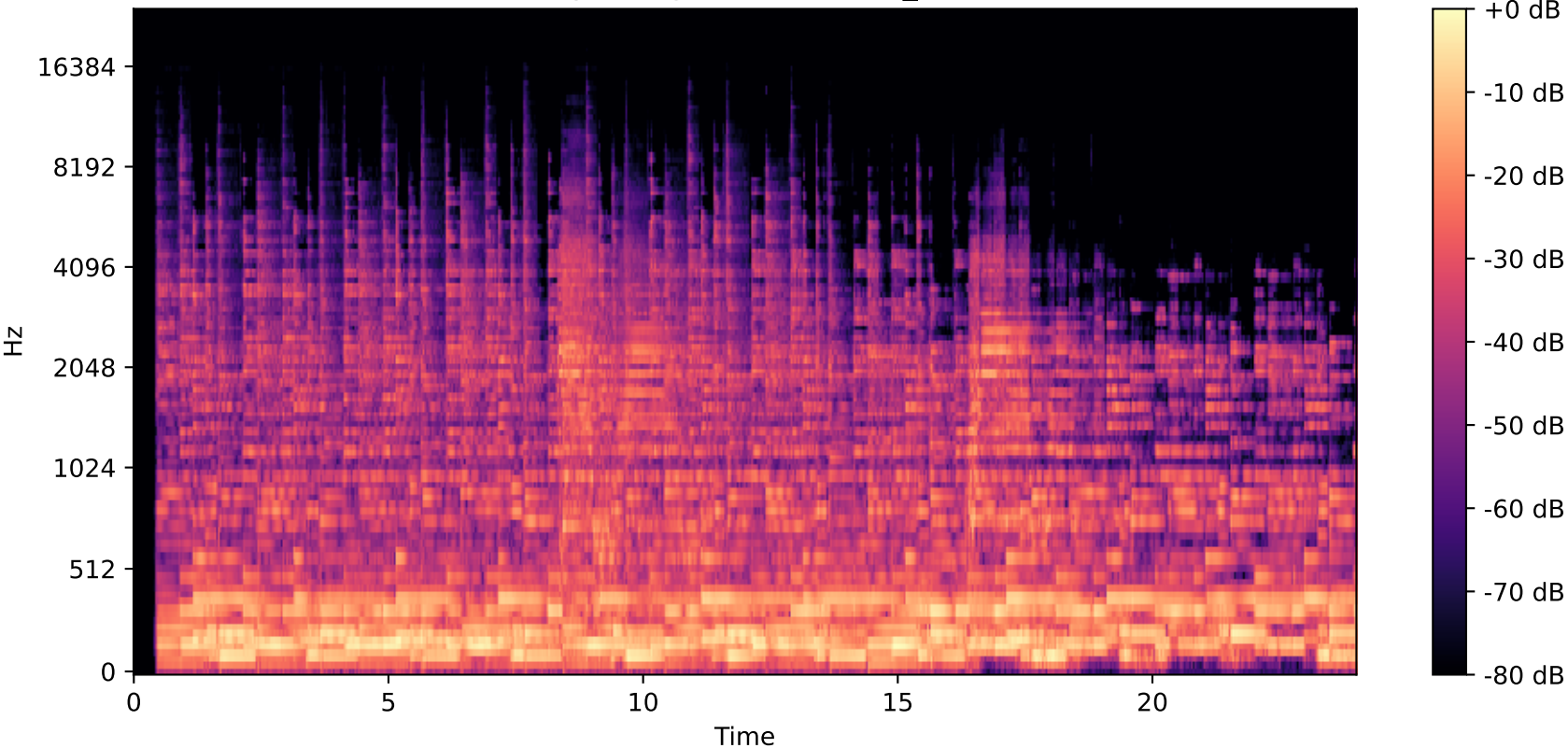
Spectrum Centroid = 862 Hz Rolloff (85%) = 398 Hz Flatness = 0.000

Distortion THD $\approx$ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 1.00)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -36.1 160 Hz: -32.8 320 Hz: -36.4 640 Hz: -44.9
1280 Hz: -44.1 2560 Hz: -49.8 5120 Hz: -60.9 10240 Hz: -82.3

Spectrogram — section_0



section_1 · 24.00 s - 86.00 s · high_gain

Labels tone_profile = high_gain dynamics_profile = rhythmic presence = lead

Loudness RMS = -23.3 dB Peak = -9.9 dB Crest = 13.3 dB

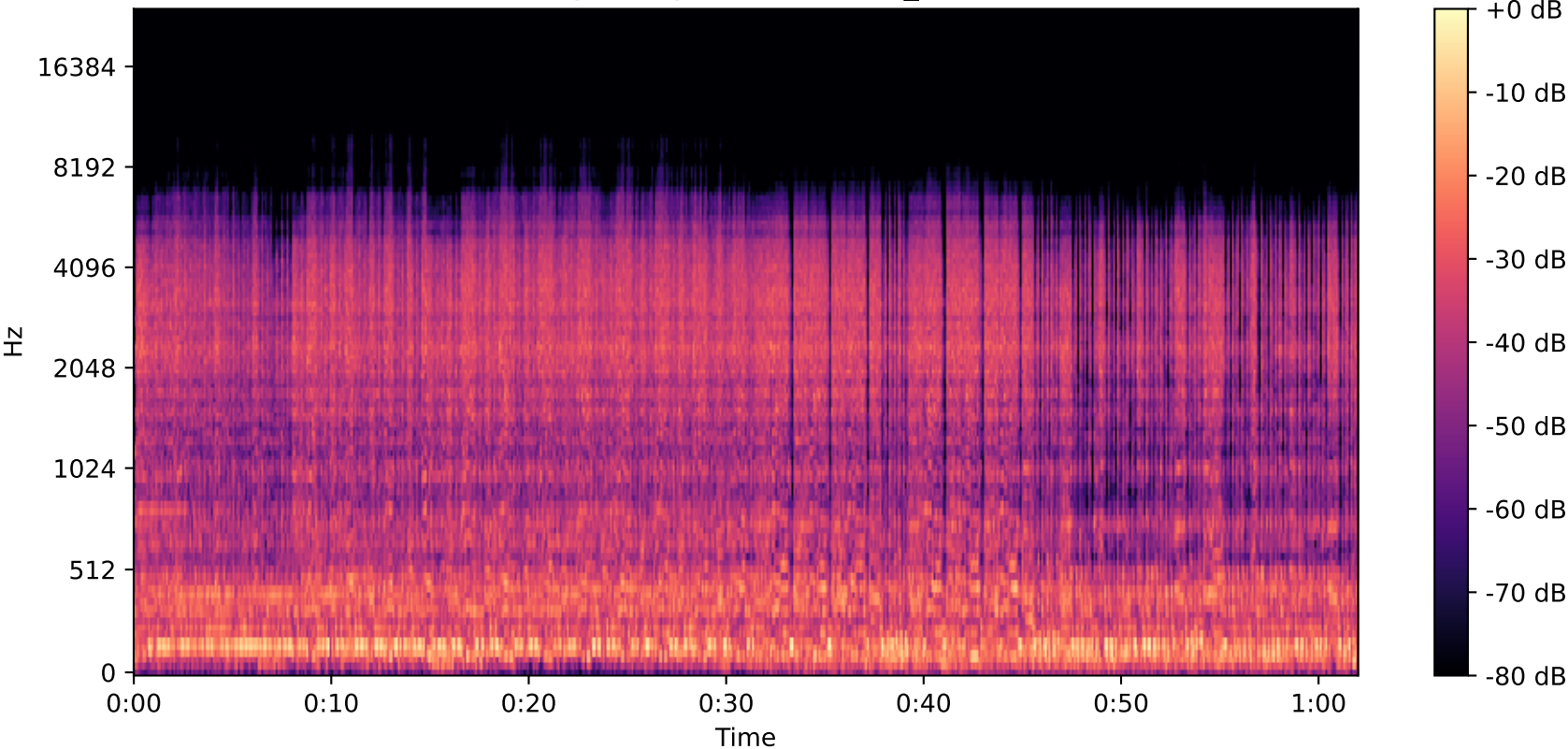
Spectrum Centroid = 2079 Hz Rolloff (85%) = 1570 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -10.1 dB gain_character = high_gain (conf 0.70)

Time FX RT60 = - s delay = off (-, fb -) mod = on (4.1 Hz, depth 1.00)

Band energy (dB) 80 Hz: -31.5 160 Hz: -27.4 320 Hz: -34.6 640 Hz: -41.5
1280 Hz: -36.3 2560 Hz: -35.0 5120 Hz: -52.2 10240 Hz: -95.7

Spectrogram — section_1



section_2 · 86.00 s - 105.00 s · distortion

Labels tone_profile = distortion dynamics_profile = rhythmic presence = rhythm

Loudness RMS = -27.0 dB Peak = -11.8 dB Crest = 15.1 dB

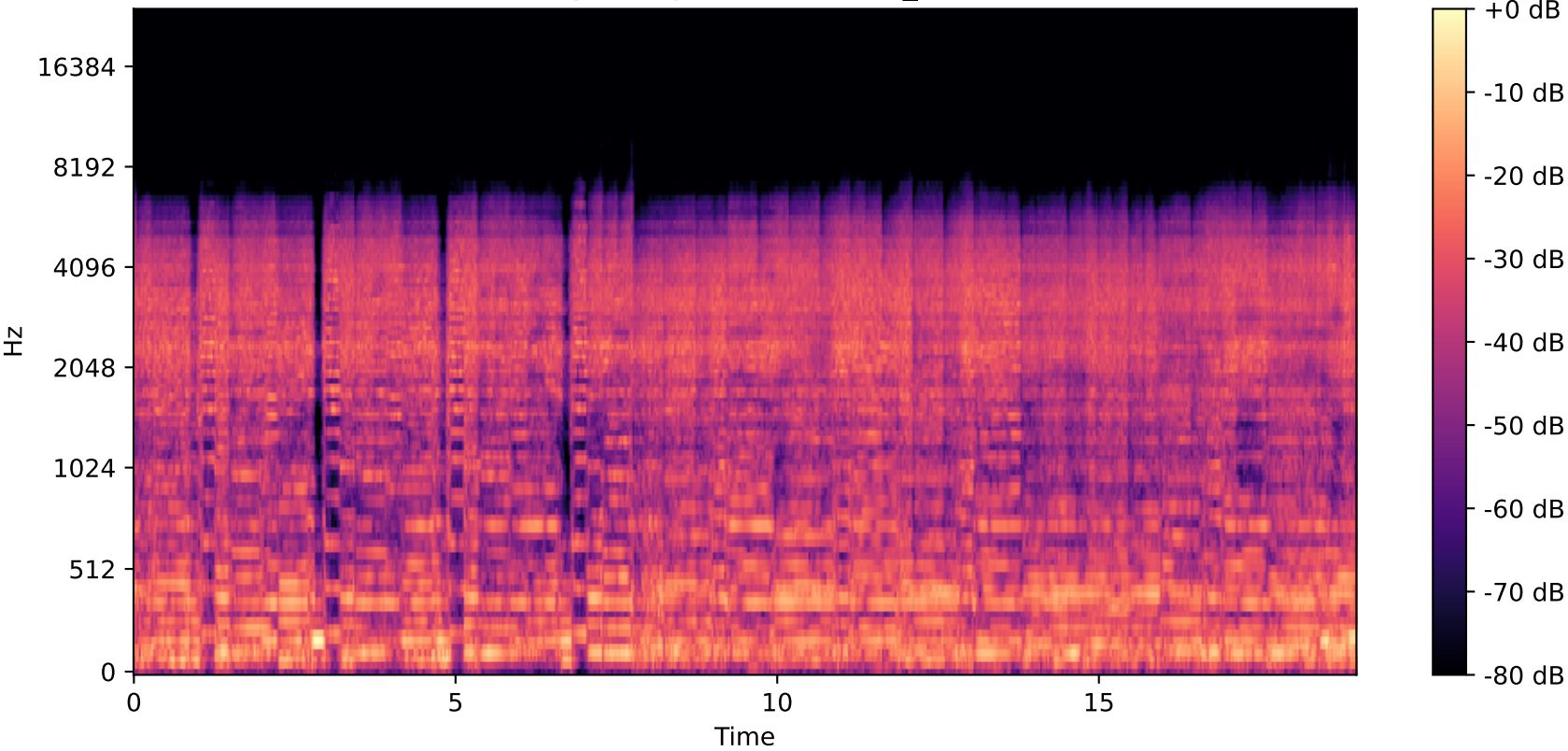
Spectrum Centroid = 2185 Hz Rolloff (85%) = 2930 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = +4.6 dB gain_character = distortion (conf 0.50)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -35.0 160 Hz: -34.9 320 Hz: -35.3 640 Hz: -41.7
1280 Hz: -36.9 2560 Hz: -35.7 5120 Hz: -53.8 10240 Hz: -104.5

Spectrogram — section_2



section_3 · 105.00 s - 138.00 s · high_gain

Labels tone_profile = high_gain dynamics_profile = rhythmic presence = lead

Loudness RMS = -23.8 dB Peak = -9.5 dB Crest = 14.3 dB

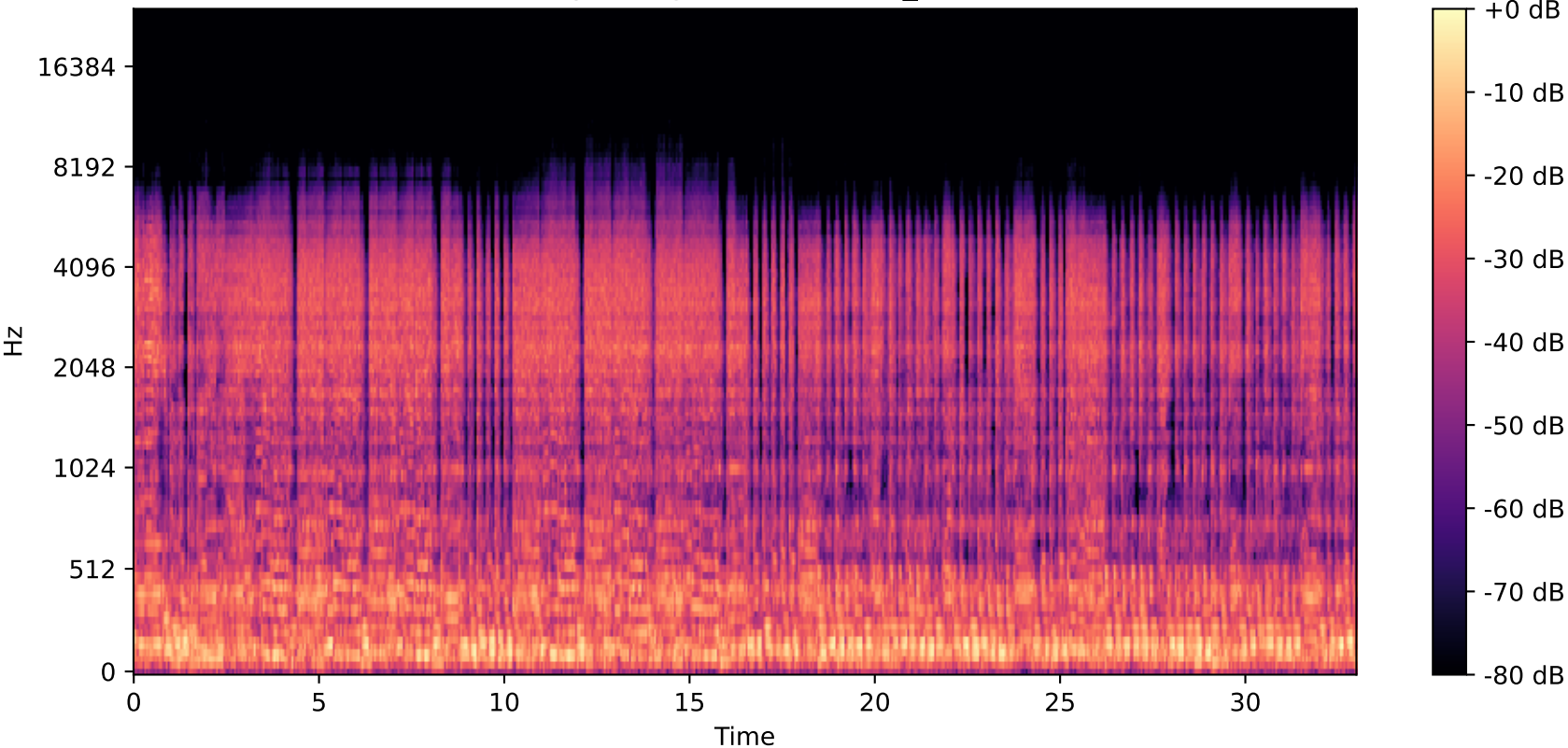
Spectrum Centroid = 2073 Hz Rolloff (85%) = 2016 Hz Flatness = 0.000

Distortion THD ≈ 45.0 % odd/even = -7.0 dB gain_character = high_gain (conf 1.00)

Time FX RT60 = - s delay = on (950 ms, fb 90%) mod = off (-, depth -)

Band energy (dB) 80 Hz: -29.7 160 Hz: -29.7 320 Hz: -34.5 640 Hz: -42.2
1280 Hz: -36.8 2560 Hz: -34.9 5120 Hz: -52.7 10240 Hz: -98.0

Spectrogram — section_3



section_4 · 138.00 s - 197.00 s · distortion

Labels tone_profile = distortion dynamics_profile = rhythmic presence = lead

Loudness RMS = -25.0 dB Peak = -9.8 dB Crest = 15.1 dB

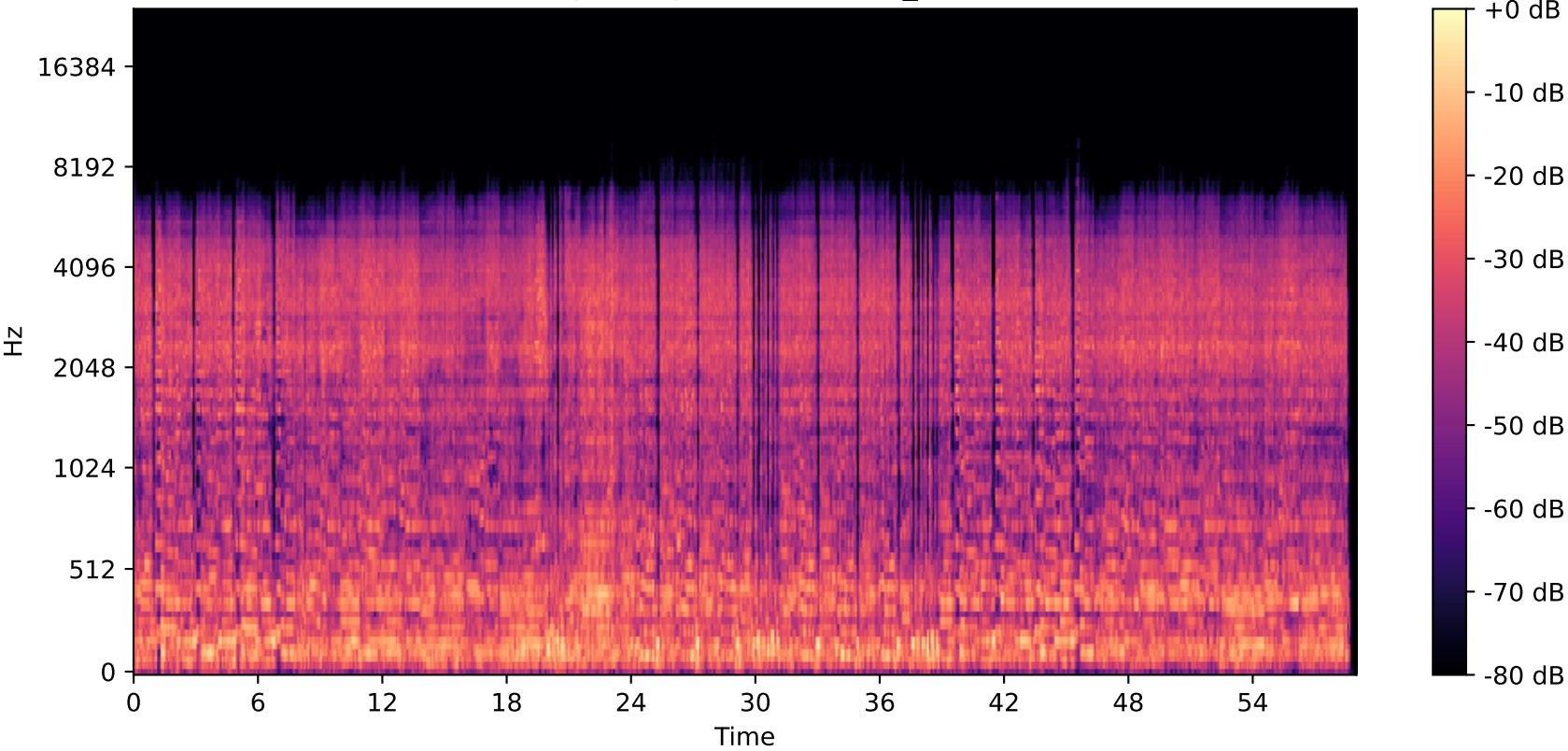
Spectrum Centroid = 2025 Hz Rolloff (85%) = 2414 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -5.3 dB gain_character = distortion (conf 0.50)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -32.5 160 Hz: -31.8 320 Hz: -33.1 640 Hz: -40.9
1280 Hz: -36.1 2560 Hz: -35.2 5120 Hz: -54.2 10240 Hz: -98.1

Spectrogram — section_4



section_5 · 197.00 s - 221.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = rhythm

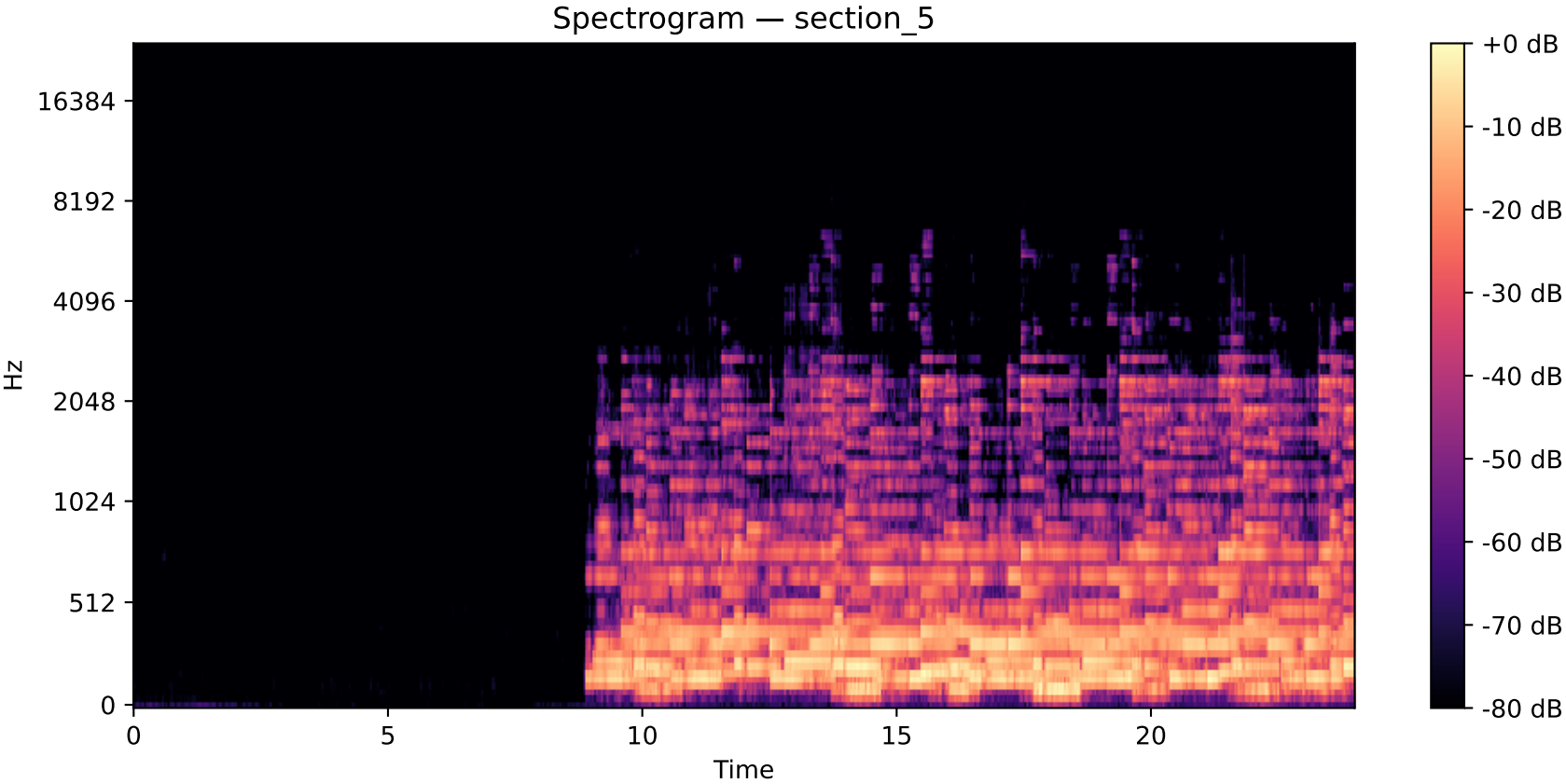
Loudness RMS = -30.2 dB Peak = -15.4 dB Crest = 14.7 dB

Spectrum Centroid = 478 Hz Rolloff (85%) = 352 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -40.7 dB gain_character = clean (conf 0.60)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -40.2 160 Hz: -34.0 320 Hz: -37.1 640 Hz: -45.7
1280 Hz: -51.3 2560 Hz: -65.9 5120 Hz: -78.4 10240 Hz: -106.8



section_6 · 221.00 s - 277.00 s · high_gain

Labels tone_profile = high_gain dynamics_profile = rhythmic presence = lead

Loudness RMS = -24.2 dB Peak = -6.6 dB Crest = 17.6 dB

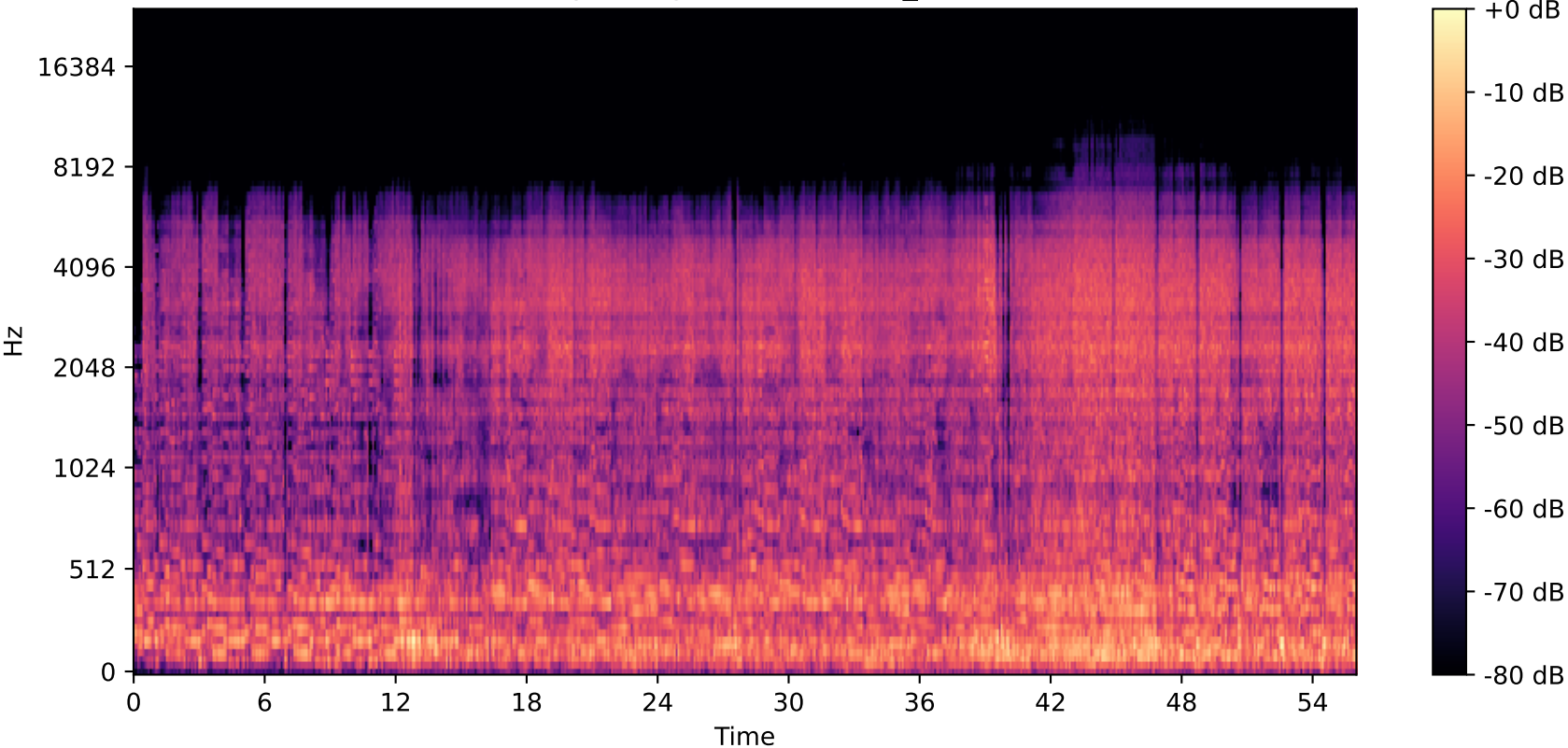
Spectrum Centroid = 2041 Hz Rolloff (85%) = 2391 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -4.9 dB gain_character = high_gain (conf 0.70)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -31.9 160 Hz: -30.7 320 Hz: -32.2 640 Hz: -40.5
1280 Hz: -35.8 2560 Hz: -34.4 5120 Hz: -50.7 10240 Hz: -90.7

Spectrogram — section_6



section_7 · 277.00 s - 309.00 s · distortion

Labels tone_profile = distortion dynamics_profile = rhythmic presence = lead

Loudness RMS = -23.5 dB Peak = -10.2 dB Crest = 13.3 dB

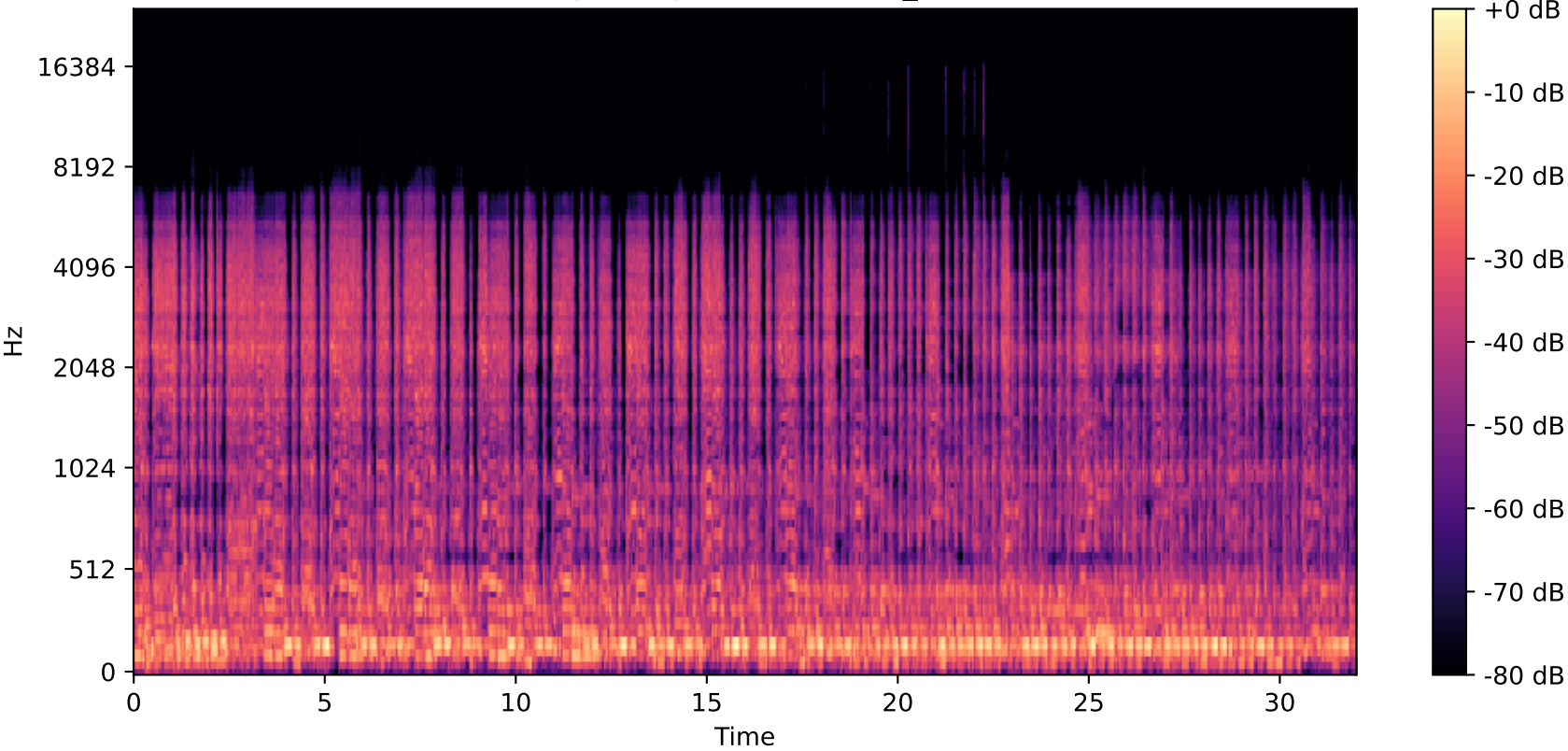
Spectrum Centroid = 1736 Hz Rolloff (85%) = 352 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -12.7 dB gain_character = distortion (conf 0.50)

Time FX RT60 = - s delay = on (300 ms, fb 100%) mod = on (4.1 Hz, depth 1.00)

Band energy (dB) 80 Hz: -32.0 160 Hz: -26.7 320 Hz: -36.8 640 Hz: -42.7
1280 Hz: -38.6 2560 Hz: -37.8 5120 Hz: -53.1 10240 Hz: -76.4

Spectrogram — section_7



section_8 · 309.00 s - 331.56 s · distortion

Labels tone_profile = distortion dynamics_profile = sparse presence = background

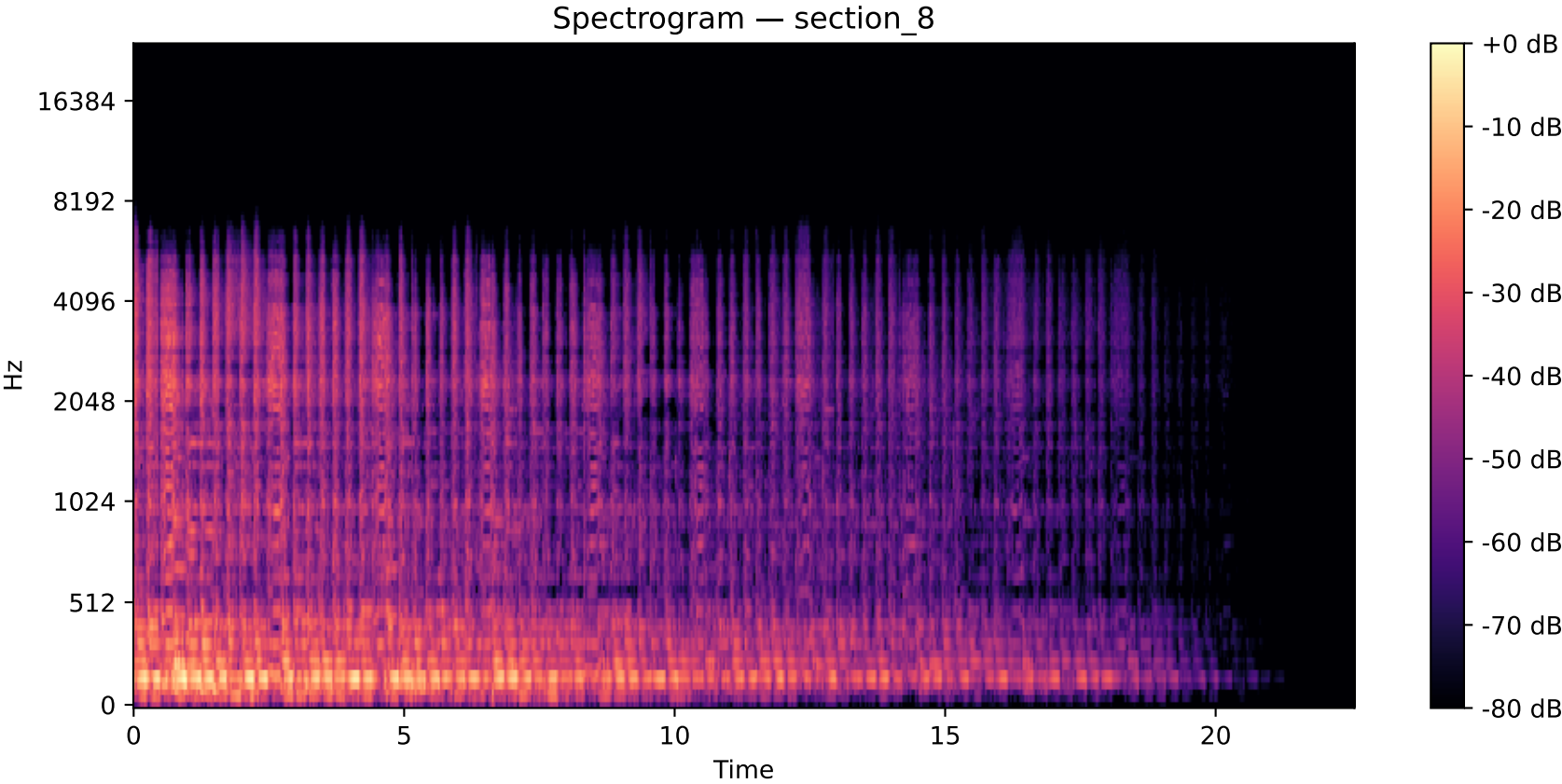
Loudness RMS = -37.4 dB Peak = -19.0 dB Crest = 18.4 dB

Spectrum Centroid = 1442 Hz Rolloff (85%) = 258 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -9.1 dB gain_character = distortion (conf 0.50)

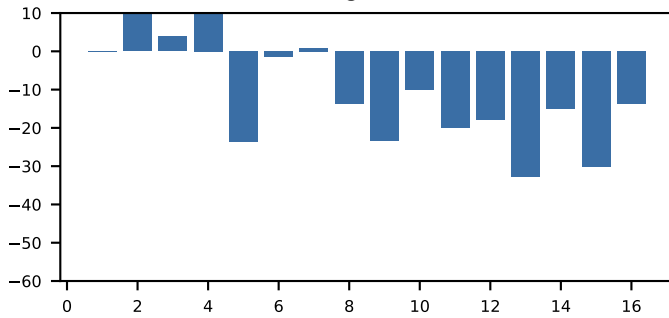
Time FX RT60 = - s delay = on (1476 ms, fb 81%) mod = on (4.1 Hz, depth 0.31)

Band energy (dB) 80 Hz: -46.4 160 Hz: -40.1 320 Hz: -51.3 640 Hz: -57.9
1280 Hz: -54.2 2560 Hz: -54.5 5120 Hz: -70.5 10240 Hz: -106.3

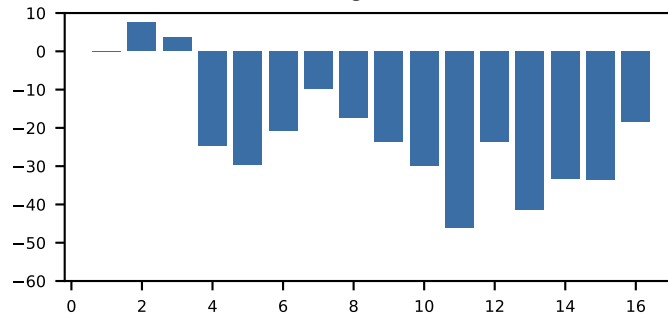


Notes

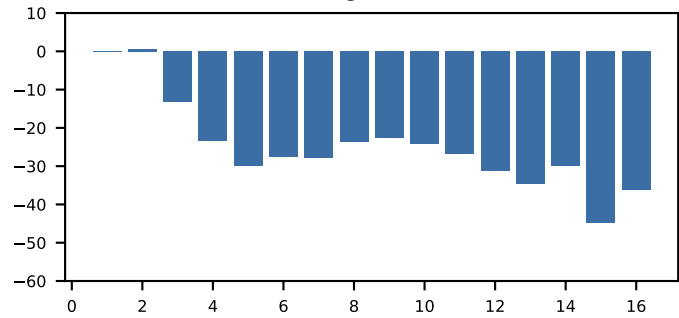
E2 @ 0.92s



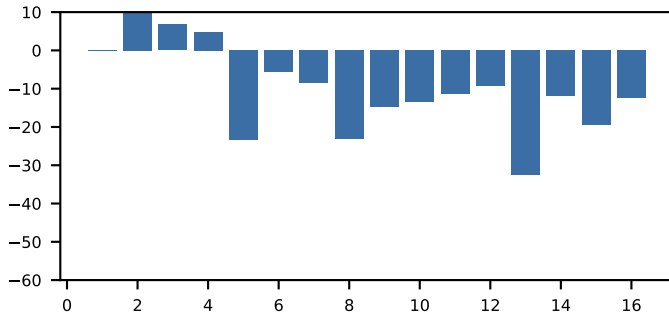
A2 @ 3.41s



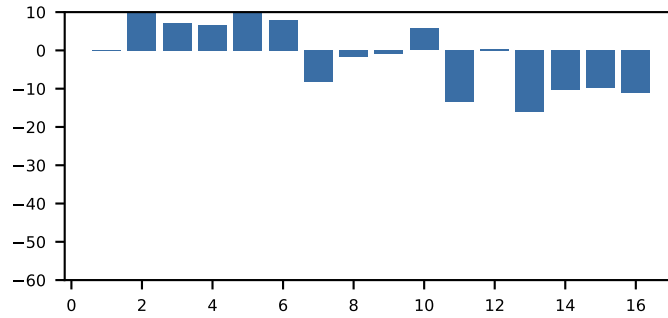
G3 @ 11.16s



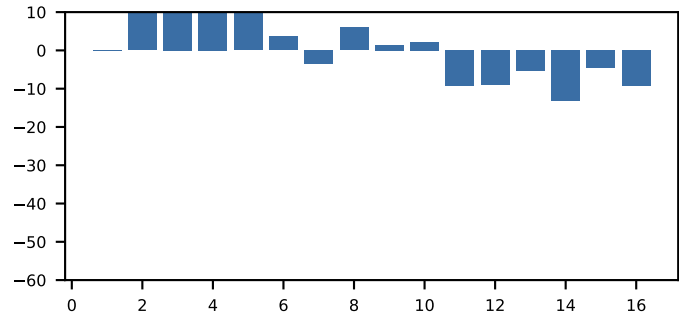
E2 @ 14.89s



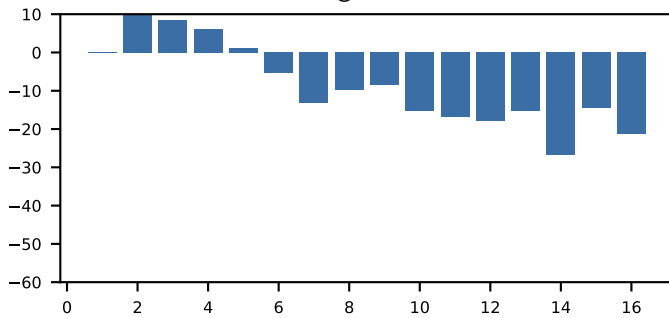
E2 @ 25.13s



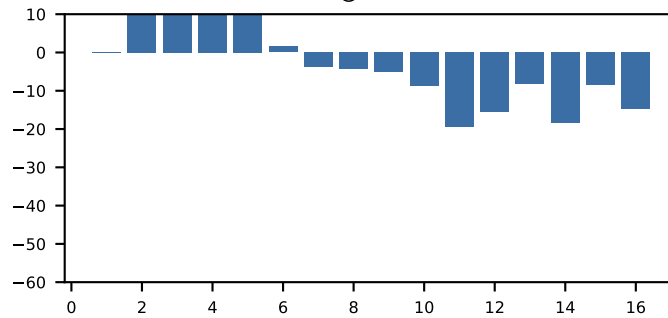
E2 @ 27.11s



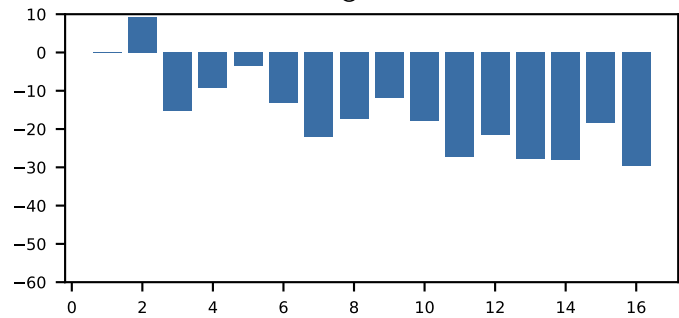
E2 @ 29.03s



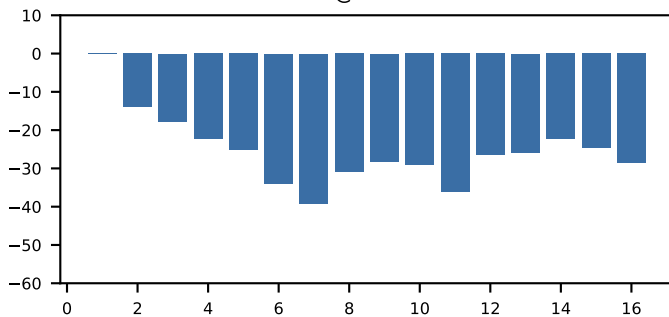
E2 @ 29.57s



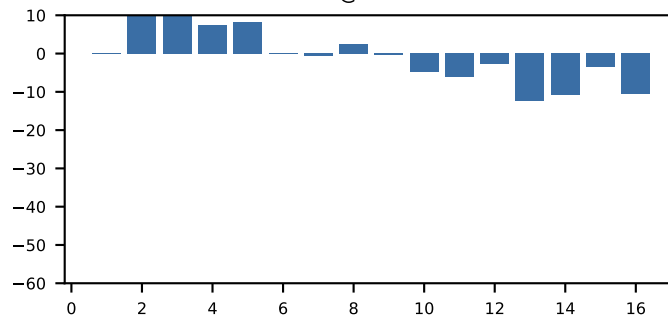
E2 @ 39.85s



E3 @ 50.86s



E2 @ 51.56s



E3 @ 75.03s

